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Topological Uniqueness of the Nash Equilibrium for Non-cooperative Routing

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Abstract

We consider the problem of selfish routing in a congested network with finitely-many users, where each user wishes to minimize the cost of its own flow. Flows may be split over different routes. The cost function is additive, with user-specific link costs that may depend on the user's own flow as well as on the total link flow. Our main interest here is network topologies that ensure the uniqueness of the Nash equilibrium under mild convexity assumptions on the link costs. We characterize the class of two-terminal network topologies for which this uniqueness property holds, and show that this class coincides with the set of *nearly parallel networks* that was recently introduced by Milchtaich [20] in the context of Wardrop equilibrium analysis. We further show how our results imply and extend known results on uniqueness of the Wardrop equilibrium in multi-class networks.

Key Words: selfish routing, congested networks, Nash equilibrium
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OR/MS subject classification: non-cooperative games; multi-commodity networks

1 Introduction

Congestion-prone networks have been an object of interest in engineering and operations research for over five decades, motivated by major applications to traffic engineering and communication networks. In these and other application domains, the network is often not centrally controlled but rather shared by a number of users who pursue their own objectives. This has led to extensive work on the analysis of multi-user networks within the framework of game theory, and to the investigation of equilibrium concepts for these models. For a recent survey on these issues from the telecommunications perspective see [2].

We consider here the problem of *competitive routing*, where each user needs to deliver a given amount of flow over the network from its designated origin node to its destination. A user can choose how to divide its flow between the available routes. On each link the user incurs a certain cost per unit flow, which in general will depend on the link congestion, namely the total flow over that link. In the context of computer networks the per-unit cost is often synonymous to the link *latency*, a terminology that we adopt here for simplicity. The latency of a path is simply the sum of the latencies along its links.

The fundamental notion of equilibrium in transportation networks has been proposed by Wardrop in [26]. Essentially, it requires all traffic to occupy paths with minimal latency. While this solution concept has been addressed by different names, including the Nash equilibrium for infinitesimal users, user equilibrium, or traffic equilibrium, we shall use the term *Wardrop equilibrium* to distinguish it from the finite-user Nash equilibrium that is the main topic of this paper. The Wardrop equilibrium arises naturally when the flow is considered to be composed of infinitesimal users, so that the effect of each user on link congestion is negligible. This equilibrium concept is also relevant in the context of computer networks, as many of the current dynamic routing protocols focus on shortest path routing. For recent overviews of the extensive literature that has been devoted to the Wardrop equilibrium and its variants see, for example, [23, 21, 2].

When cost-optimizing users control non-negligible amount of flow, we are led to consider the standard Nash equilibrium for finitely-many users. A flow pattern is considered to be a Nash equilibrium point (NEP) if no user can reduce the total cost incurred by its traffic by a unilateral change of its own flow pattern. While this problem has more recent roots than the Wardrop equilibrium, it has attracted considerable attention in recent literature. Perhaps the first paper to study this problem is [13], which shows convergence of the Nash equilibrium to

the Wardrop equilibrium as the number of users increases to infinity. Existence, uniqueness and some basic properties of the Nash equilibrium are studied in [22, 1, 3]. The notion of a mixed Nash-Wardrop equilibrium, which combines infinitesimal users with finitely-sized ones, is considered in [12, 6]. Efficient network design and management are considered in [15, 16, 18, 17, 11], while [25] bounds the performance degradation relative to centralized routing (along with similar results for the Wardrop equilibrium). The convergence of some dynamic schemes to the Nash equilibrium is considered in [14], while [19] considers a repeated game version of the routing problem, and [4] considers the addition of side-constraints on the flows.

We focus here on the issue of uniqueness of the Nash equilibrium in noncooperative routing with finitely-manu users. For a two-node network with parallel links, uniqueness of the Nash equilibrium has been established under mild convexity assumptions on the link costs [22]. This result does not hold for networks of general topology, as demonstrated there by a counter-example. However, the question of whether there exist other network topologies for which uniqueness of the Nash equilibrium is guaranteed (under similar convexity assumptions) remained open until now.

For networks of general topology, the uniqueness of the Nash equilibrium has been established under various additional conditions on the cost functions. A general set of conditions is implied by the requirement of diagonal strict convexity, which is a well known sufficient condition for uniqueness of the Nash equilibrium in convex (or concave) games [24]. These conditions have been applied to the Nash routing problem in [13, 22]. Unfortunately, those conditions do not hold in many cases of interest – for example, they are violated by popular M/M/1 latency function under significant congestion. Other special cases are presented in [1] and [3]: the first considers link latencies that are polynomial with a low enough order, while the latter establishes uniqueness under some specific symmetry conditions.

For the Wardrop equilibrium, a corresponding line of uniqueness results exists, with the requirement of link cost convexity replaced by monotonicity of the link latency. This is sufficient to guarantee uniqueness in the single-class case, but not for the multi-class problem [9, 8]. Additional conditions on the costs that ensure uniqueness are considered in [7, 8, 3]. In a recent paper, Milchtaich [20] provides a complete characterization of all two-terminal network topologies (called *nearly parallel networks*) for which uniqueness is guaranteed under the basic monotonicity requirement. These results are highly relevant for the present paper and will be further elaborated in the sequel.

Our goal here is to characterize those network topologies for which the *Nash equilibrium* is unique, for any number and size of users, as long as their link cost functions satisfy some mild convexity conditions. Our main results establish that the class of networks that satisfy this property coincides with the set of nearly-parallel networks. We shall also briefly elaborate on the relation between the Wardrop and Nash equilibrium, and show that, in a sense, our results imply and extend the uniqueness results for the former.

The paper is organized as follows: In section 2 we present the model, and repeat the definition of nearly parallel networks from [20]. Section 3 establishes our main results concerning uniqueness of the Nash equilibrium, while Section 4 discusses the relation between the Nash and Wardrop equilibria. Finally, Section 5 offers some concluding remarks.

2 Model and Preliminaries

2.1 The Network Model

Let the network topology be specified by an undirected graph $\mathcal{G} = \mathcal{G}(\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is a finite set of vertices (or *nodes*) and $\mathcal{E}$ is a finite set of edges. Each edge joins two distinct vertices. Thus, single-edge loops are not allowed, but more than one edge can join two vertices. Two of the vertices in this graph will be designated as the *terminal* vertices, O (for origin) and D (for destination). We refer to such a graph as a *two-terminal network*.

The actual network we consider is directional, and is obtained from the undirected graph by replacing each edge with two directional links. Thus, an edge e between vertices u and v is split into two directional links, one from u to v and the other from v to u. (We shall comment about that particular way of defining the directional network at the end of section 3. For now, note that by imposing large enough costs on some of the links we may be effectively obtain any subnetwork thereof.) The set of links that connect u to v is denoted by $\mathcal{L}(u, v)$ and the set of all links in the directional network by $\mathcal{L}$.

We are given a set $\mathcal{I} = \{1, 2, \ldots, I\}$ of users, which share the network. Each user i needs to deliver a positive amount d^i of flow from node O to node D, and should decide how to divide its flow between the different routes that connect these two nodes. Denote by f_l^i the flow of user i through link l, and let $f_l = \sum_{i \in \mathcal{I}} f_l^i$ denote the total flow on link l. User i's flow pattern is the vector $f^i = (f_l^i,\ l \in \mathcal{L})$. Finally, the system flow pattern f is the vector

of all user flow patterns.

A feasible flow pattern must obey the following flow conservation and positivity constraints for each user:

$$\sum_{l \in OUT(v)} f_l^i = \sum_{l \in IN(v)} f_l^i + d_v^i, \quad v \in \mathcal{V} \tag{1}$$

$$f_l^i \geq 0, \quad l \in \mathcal{L}\,. \tag{2}$$

Here $In(v)$ and $Out(v)$ are the set of input and output links to node v, and

$$d_v^i = \begin{cases} d^i & : \quad v = O \\ -d^i & : \quad v = D \\ 0 & : \quad v \neq O, D \end{cases} \tag{3}$$

We denote the set of feasible flow patterns f^i for user i by F^i. This is clearly a convex polyhedron. A system flow pattern f is feasible if $f^i \in F^i$ for all $i \in \mathcal{I}$.

2.2 Convex Network Games

The performance measure to be minimized by user $i \in \mathcal{I}$ is specified by a cost function $J^i(f)$. We shall consider additive cost functions of the form

$$J^i(f) = \sum_{l \in \mathcal{L}} J_l^i(f_l^i, f_l)\,. \tag{4}$$

Thus, the cost incurred by a user on a link l depends only on its own flow f_l^i, as well as on the total link flow f_l which measures the link congestion. Link costs are often taken to be in the form $J_l^i(f_l^i, f_l) = f_l^i\, T_l^i(f_l)$, where $T_l^i(f_l)$ represents the cost per unit flow (or *latency*). Note that the link cost functions J_l^i may depend on the user i, and similarly for the latency T_l^i.

Definition 1 *A flow vector f_* is a Nash equilibrium point (NEP) if, for each $i \in \mathcal{I}$,*

$$J^i(f_*) = \min_{f^i \in F^i} J^i(f_*^1, \ldots, f_*^{i-1}, f^i, f_*^{i+1}, \ldots, f_*^I). \tag{5}$$

Denote by $\mathcal{J}$ the vector of link costs functions $(J_l^i,\ l \in \mathcal{L}, i \in \mathcal{I})$. Also denote by d the vector $(d^1, \ldots, d^I)$ which specifies the demand of all users. Let

$$K_l^i(f_l^i, f_l) \stackrel{\triangle}{=} \frac{\partial J_l^i(f_l^i, f_l)}{\partial f_l^i} + \frac{\partial J_l^i(f_l^i, f_l)}{\partial f_l}$$

denote the marginal costs of user i on link l with respect to its flow f_l^i (where the second term is required since $f_l = \sum_i f_l^i$). We shall assume that the link cost functions satisfy the following properties.

Assumption A1: J_l^i is a continuous, non-negative function on its domain $\{(x,y) \in \Re_+^2 : x \le y\}$.

Assumption A2: J_l^i is strictly increasing in each of its arguments.

Assumption A3: J_l^i is continuously differentiable, and the marginal cost $K_l^i(f_l^i, f_l)$ is strictly increasing in each of its arguments.

Functions that satisfy these assumptions were termed *type-A cost functions* in [22]. We note that in this reference the costs were allowed to take infinite values, in order to accommodate popular cost functions such as the M/M/1 latency function: $J_l^i(f_l^i, f_l) = \frac{f_l^i}{c_l - f_l}$, with $J_l^i = \infty$ for $f_l \ge c_l$. This extension is easily accommodated in the present paper, with all uniqueness results directly applicable to finite NEPs (namely, NEPs where all users incur finite costs). The existence of NEPs with infinite costs should then be excluded by other assumptions and considerations.

Definition 2 *A* convex network game *over a two-terminal network $\mathcal{G}$ is a triple $(\mathcal{I}, d, \mathcal{J})$ over $\mathcal{G}$, with cost functions that satisfy Assumptions A1-A3 for each link l and user i.*

For a convex network game, the minimization problem faced by each user (with the flow pattern of the others held fixed) is a convex optimization problem, as convexity of the link costs in each user's flow is implied by our Assumption A3. Necessary and sufficient conditions for a flow pattern to be a Nash equilibrium are therefore provided by the Karush-Kuhn-Tucker conditions, applied to each user in turn [22]: A flow pattern f is a Nash equilibrium if there exists a set of positive constants $\{\lambda_u^i : \ i \in \mathcal{I}, \ u \in \mathcal{V}\}$ so that for every link $l \in \mathcal{L}(u,v)$, and for every user i:

$$\lambda_u^i = K_l^i(f_l^i, f_l) + \lambda_v^i \qquad \text{if } \ f_l^i > 0\,, \tag{6}$$

$$\lambda_u^i \le K_l^i(f_l^i, f_l) + \lambda_v^i \qquad \text{if } \ f_l^i = 0\,. \tag{7}$$

Conditions (6) and (7) can also be expressed in the following path-oriented manner: For any two nodes in the network u and v, and any path p that connects u and v, if $f_l^i > 0$ for every $l \in p$ then

$$\lambda_{uv} \triangleq \lambda_u - \lambda_v = \sum_{l \in p} K_l^i(f_l^i, f_l) \le \sum_{l \in p'} K_l^i(f_l^i, f_l) \tag{8}$$

where p' is any other path connecting u and v.

In general, the NEP of a convex network game need not be unique. Still, uniqueness is guaranteed for certain network topologies. We shall refer to this property as *topological uniqueness*. More precisely:

Definition 3 *A network $\mathcal{G}$ has the* topological uniqueness property *if the NEP is unique for any convex network game over $\mathcal{G}$.*

2.3 Nearly Parallel Networks

We briefly repeat here some definitions and results from [20] concerning nearly-parallel networks. These results show that network topologies can be divided into two classes. The first essentially contains the networks shown in Figure 1, and networks which are serial connection of those networks. The second class contains all networks in which one of the basic networks shown in Figure 2 is embedded, in the following sense.

Definition 4 *A network $\mathcal{G}'$ is said to be embedded in the wide sense in network $\mathcal{G}''$ if the latter can be obtained from the former by some sequence of the following three operations:*

1. Edge subdivision: *An edge is replaced by two edges with a single common end vertex.*
2. Edge addition: *The addition of a new edge joining two existing vertices.*
3. Terminal vertex subdivision: *The addition of a new edge, joining the terminal vertex O or D with a new vertex v, such that a nonempty subset of the edges originally incident with the terminal vertex are incident with v instead.*

Definition 5 *A two-terminal network $\mathcal{G}$ is called* nearly parallel *if it is one of the networks in Figure 1, or can be constructed from one of the networks in Figure 1 by a series of edge subdivisions.*

Of the five networks in Figure 1 network (e) is the most interesting, as the other four may be considered a special case of this network for routing purposes. However, the formal definition of nearly parallel networks does require to consider all these basic networks. Note also that only network (e) supports meaningful bi-directional traffic (between nodes A and B) when routing traffic between the prescribed source and destination nodes.

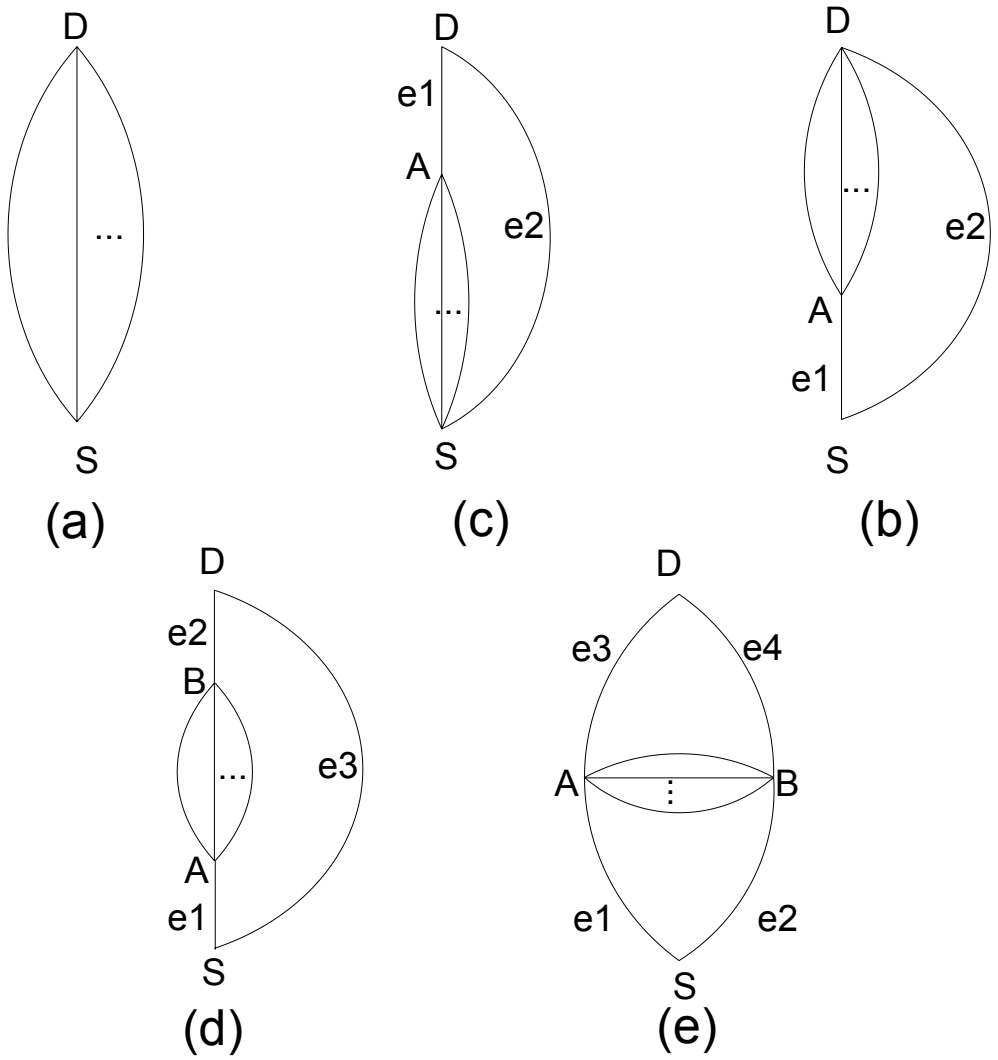


Figure 1: Basic networks that define the class of nearly-parallel networks

Proposition 1 ([20]) *For every two-terminal network $\mathcal{G}$, one, and only one, of the following conditions holds:*

(i) $\mathcal{G}$ is nearly parallel or is a serial connection of two or more nearly parallel networks.

(ii) One (or more) of the networks in Figure 2 is embedded in the wide sense in $\mathcal{G}$.

To simplify terminology, from here on we shall use the term "nearly parallel network" to refer to any network that meets condition (i) of the last proposition, namely both to nearly parallel networks in the sense of Definition 5 *and* to serial connections thereof.

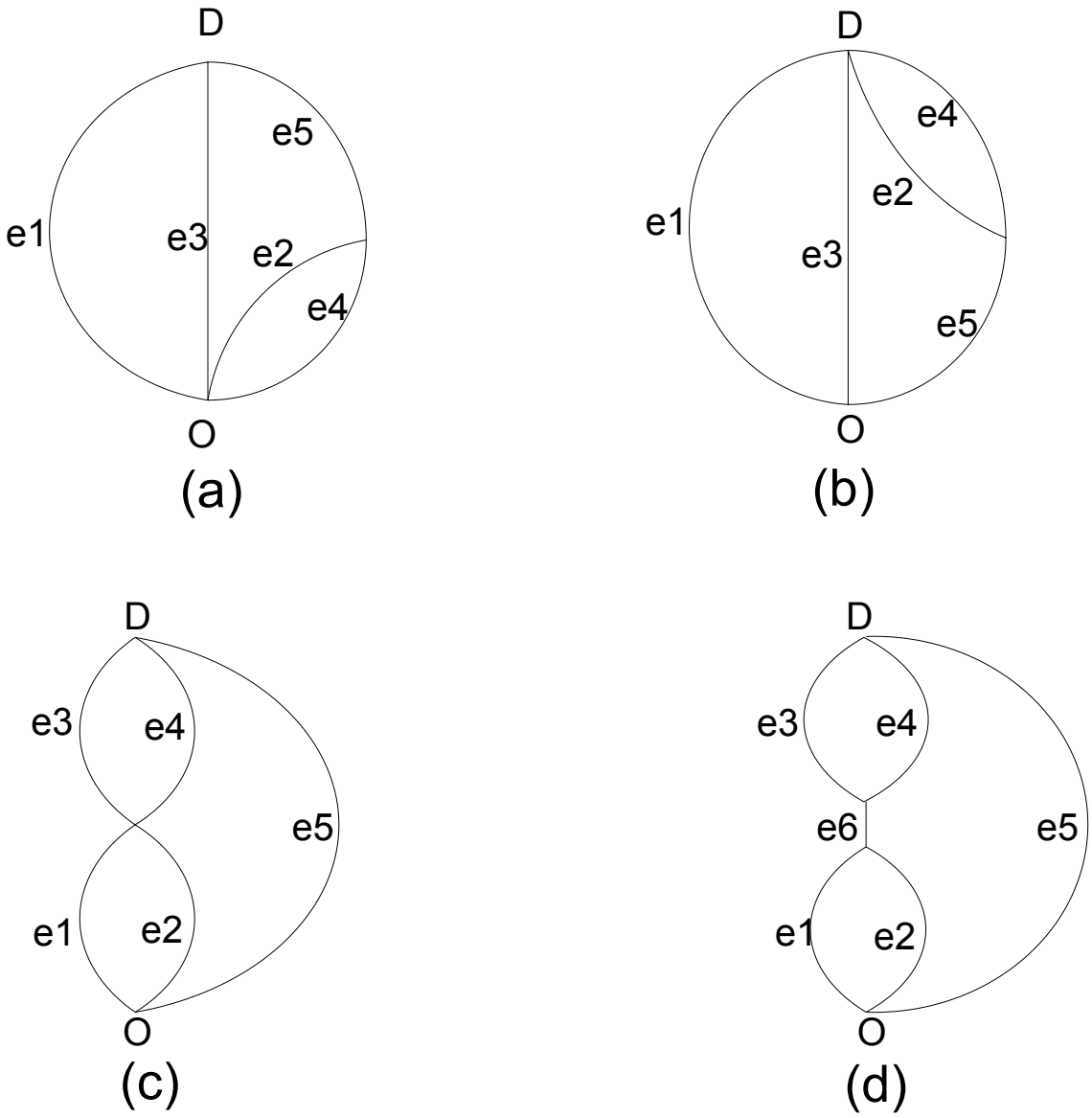


Figure 2: Basic networks that are not nearly-parallel

3 Uniqueness of the Nash Equilibrium

It has been established in [22] that parallel-link networks posses the topological uniqueness property, namely that uniqueness of the NEP is guaranteed under Assumptions A1-A3. Our main result states that topological uniqueness (in the sense of Definition 3) holds for the larger class of nearly parallel networks, and *only* for that class.

Theorem 1 *A two-terminal network $\mathcal{G}$ has the topological uniqueness property if, and only if, $\mathcal{G}$ is a nearly parallel network.*

The proof is presented in the following subsections. The next subsection establishes some monotonicity properties that hold in parallel-link networks and will be required later. Subsection 3.2 establishes uniqueness of the NEP in nearly parallel networks. Finally, subsection 3.3 shows that a counter example to uniqueness can be constructed for every network which is not nearly parallel.

3.1 Monotonicity Results for Parallel-Link Networks

A *parallel-link network* contains only two nodes, as in Figure 1(a). In the following we shall require some properties of the equilibrium flows in this network, when it considered as a part of a larger network (in particular, part of the network of Figure 1(e)). Accordingly, we shall use the symbols A and B to denote the terminals of this network. Obviously, these results hold also for the network from B to A, as well as for any stand-alone parallel-link network (with the total flows f^i_{AB} standing for the flow requirements d^i).

Consider a parallel-link network G' (possibly a part of a larger network G) which connects node A to node B. Denote by $\mathcal{L}(A,B)$ the set of links in G'. Let

$$f^i_{AB} = \sum_{l \in \mathcal{L}(A,B)} f^i_l \tag{9}$$

denote the total flow of user i on that network. Let f be an equilibrium flow in G. The equilibrium conditions (6) and (7) imply that for every $l \in \mathcal{L}(A,B)$,

$$K^i_l(f^i_l, f_l) = \lambda^i_{AB} \qquad \text{if} \quad f^i_l > 0 \tag{10}$$

$$K^i_l(f^i_l, f_l) > \lambda^i_{AB} \qquad \text{if} \quad f^i_l = 0 \tag{11}$$

where $\lambda^i_{AB} = \lambda^i_B - \lambda^i_A$.

Consider two Nash equilibria of G, denoted f and $\hat{f}$. Throughout this proof hatted symbols designate quantities related to $\hat{f}$, while plain symbols refer to the NEP f.

Lemma 1 *If $\hat{f}_l \geq f_l$ and $\hat{\lambda}^i \leq \lambda^i$, then $\hat{f}^i_l \leq f^i_l$.*

Proof: Assume $\hat{f}_l \geq f_l$ and $\hat{\lambda}^i \leq \lambda^i$. The KKT conditions state that:

$$\text{either} \quad \hat{f}_l^i = 0 \leq f_l^i, \quad \text{or} \quad K_l^i(\hat{f}_l^i, \hat{f}_l) = \hat{\lambda}^i \leq \lambda^i \leq K_l^i(f_l^i, f_l). \tag{12}$$

Since $\hat{f}_l \geq f_l$ and K_l^i is increasing in both arguments it follows that $\hat{f}_l^i \leq f_l^i$ in either case, which is what we set out to show. □

Proposition 2 *Let f and $\hat{f}$ be two NEPs. Let λ_{AB} and $\hat{\lambda}_{AB}$ be the corresponding Lagrange multipliers, and denote $I_{AB}^+ = \{i | \hat{\lambda}^i > \lambda^i\}$. Then $\sum_{i \in I_{AB}^+} \hat{f}_{AB}^i - f_{AB}^i \geq 0$.*

Proof: Divide the users into two distinct subsets:

$$I_{AB}^+ = \{i \in \mathcal{I} | \hat{\lambda}^i > \lambda^i\} \tag{13}$$

$$I_{AB}^- = \{i \in \mathcal{I} | \hat{\lambda}^i \leq \lambda^i\}. \tag{14}$$

Also divide the set of links $\mathcal{L}(A, B)$ into two distinct subsets

$$L_{AB}^+ = \{l \in \mathcal{L}(A, B) | \hat{f}_l > f_l\} \tag{15}$$

$$L_{AB}^- = \{l \in \mathcal{L}(A, B) | \hat{f}_l \leq f_l\}. \tag{16}$$

From Lemma 1, if $l \in L_{AB}^+$, then for every $i \in I_{AB}^-$ it holds that $\hat{f}_l^i \leq f_l^i$. Therefore

$$\sum_{i \in I_{AB}^-} \hat{f}_l^i \leq \sum_{i \in I_{AB}^-} f_l^i \,. \tag{17}$$

But since $\hat{f}_l > f_l$ for $l \in L_{AB}^+$, this implies

$$\sum_{i \in I_{AB}^+} \hat{f}_l^i > \sum_{i \in I_{AB}^+} f_l^i \,. \tag{18}$$

Summing up over L_{AB}^+ gives:

$$\sum_{l \in L_{AB}^+} \sum_{i \in I_{AB}^+} \hat{f}_l^i \geq \sum_{l \in L_{AB}^+} \sum_{i \in I_{AB}^+} f_l^i \,, \tag{19}$$

where only a weak inequality holds since L_{AB}^+ may be empty.

Let l be a link in L_{AB}^-. Then for every $i \in I_{AB}^+$,

$$K_l^i(\hat{f}_l^i, \hat{f}_l) \geq \hat{\lambda}^i > \lambda^i = K_l^i(f_l^i, f_l), \quad \text{or} \quad \hat{f}_l^i \geq f_l^i = 0. \tag{20}$$

Since $\hat{f}_l \leq f_l$ and K_l^i is strictly increasing in both arguments, it follows that $\hat{f}_l^i \geq f_l^i$. Therefore,

$$\sum_{i \in I_{AB}^+} \hat{f}_l^i \geq \sum_{i \in I_{AB}^+} f_l^i \,. \tag{21}$$

Summing up over L_{AB}^- now gives:

$$\sum_{l \in L_{AB}^-} \sum_{i \in I_{AB}^+} \hat{f}_l^i \geq \sum_{l \in L_{AB}^-} \sum_{i \in I_{AB}^+} f_l^i \,. \tag{22}$$

Combining (22) and (19) yields:

$$\sum_{i \in I_{AB}^+} \hat{f}_{AB}^i = \sum_{l \in \mathcal{L}(A,B)} \sum_{i \in I_{AB}^+} \hat{f}_l^i \geq \sum_{l \in \mathcal{L}(A,B)} \sum_{i \in I_{AB}^+} f_l^i = \sum_{i \in I_{AB}^+} f_{AB}^i \tag{23}$$

and the proposition is proven. □

Note that if L_{AB}^+ is not empty then (19) holds with a strict inequality and $\sum_{i \in I_{AB}^+} (\hat{f}_{AB}^i - f_{AB}^i) > 0$. The next proposition follows.

Proposition 3 *If $\sum_{i \in \mathcal{I}} \hat{f}_{AB}^i \geq \sum_{i \in \mathcal{I}} f_{AB}^i$ and L_{AB}^+ is not empty, then there is at least one user i for which $\hat{f}_{AB}^i > f_{AB}^i$ and $\hat{\lambda}^i > \lambda^i$.*

Proof: If L_{AB}^+ is not empty then, from Proposition 2, $\sum_{i \in I_{AB}^+} (\hat{f}_{AB}^i - f_{AB}^i) > 0$ and for at least one user $i \in I_{AB}^+$ it holds that $\hat{f}_{AB}^i > f_{AB}^i$. It remains to consider the case where there is no link in which $\hat{f}_l > f_l$, namely $\hat{f}_l \leq f_l$ for all $l \in \mathcal{L}(A,B)$. Since $\sum_{i \in \mathcal{I}} \hat{f}_{AB}^i \geq \sum_{i \in \mathcal{I}} f_{AB}^i$, this immediately implies that $\hat{f}_l = f_l$ for every $l \in \mathcal{L}(A,B)$, and consequently $\sum_{i \in \mathcal{I}} \hat{f}_{AB}^i = \sum_{i \in \mathcal{I}} f_{AB}^i$. Now, $\hat{f}_{AB}^i > f_{AB}^i$ by assumption for at least one user i, so that $\hat{f}_l^i > f_l^i$ on some link. Since $\hat{f}_l = f_l$, the monotonicity of K_l^i (from Assumption A3) together with the equilibrium conditions imply that $\hat{\lambda}^i = K_l^i(\hat{f}_l^i, \hat{f}_l) > K_l^i(f_l^i, f_l) \geq \lambda^i$. Hence the proof is complete. □

3.2 Uniqueness for Nearly Parallel Networks

We now turn to establish uniqueness of the Nash equilibrium for the class of nearly parallel networks. This subsection contains the proof of the following claim.

Proposition 4 *If $\mathcal{G}$ is nearly parallel, then the NEP is unique for every convex network game $(I, d, \mathcal{J})$ over $\mathcal{G}$.*

Proof: If $\mathcal{G}$ is a serial connection of the basic networks in Figure 1 then uniqueness in $\mathcal{G}$ is clearly equivalent to uniqueness in each one of the basic networks. Also, subdividing an edge should not change the property of topological uniqueness since each divided edge is equivalent to a single edge with the sum of the costs of its parts (obviously Assumptions A1-A3 carry over). Hence, we need only show that the Nash equilibrium for each of the networks in Figure 1 is unique. Furthermore, it may be seen that networks (a) to (c) are subnetworks of (e) so that multiple equilibria in (a) to (c) may be easily induced in (e) by imposing large enough costs on its additional links. Similarly, network (d) may be reduced to (b) or (c) by unifying links $e1$ and $e2$. It is therefore sufficient to establish uniqueness for the network in Figure 1(e), on which we focus hereafter. In this network there are two parallel-link sub-networks, one from A to B and the other from B to A. Denote the flows on these subnetworks as follows:

$$f^i_{AB} = \sum_{l\in\mathcal{L}(A,B)} f^i_l \tag{24}$$

$$f^i_{BA} = \sum_{l\in\mathcal{L}(B,A)} f^i_l \tag{25}$$

$$f_{AB} = \sum_i f^i_{AB} \tag{26}$$

$$f_{BA} = \sum_i f^i_{BA} \tag{27}$$

Note that $f^i_{AB} > 0$ and $f^i_{BA} > 0$ cannot both hold in equilibrium, since the Nash equilibrium cannot contain routing loops (cancelling those loops will always benefit the user as link costs are positive by assumption).

Let f and $\hat{f}$ denote two NEPs in network (e). We would like to show that $f = \hat{f}$. Denote $\Delta f^i_l = \hat{f}^i_l - f^i_l$, and similarly $\Delta f^i_{AB} = \hat{f}^i_{AB} - f^i_{AB}$ etc. Flow conservation yields the following relations (which obviously hold also for the total link flows):

$$\Delta f^i_{e1} = -\Delta f^i_{e2} \tag{28}$$

$$\Delta f^i_{e3} = -\Delta f^i_{e4} \tag{29}$$

$$\Delta f^i_{e1} + \Delta f^i_{BA} = \Delta f^i_{AB} + \Delta f^i_{e3} \tag{30}$$

The next Lemma contains several elementary claims that are needed later on, and follows directly from the monotonicity properties of the marginal costs.

Lemma 2 *For the network in Figure 1(e), let f and $\hat{f}$ be two NEPs and λ and $\hat{\lambda}$ be the corresponding Lagrange multiplier vectors:*

a) If $\Delta f_{e1} \geq 0$, $\Delta f_{e3} \geq 0$ and $\Delta f^i_{e3} > 0$ then $\Delta f^i_{e1} \leq 0$.

b) If $\Delta f_{e1} \geq 0$, $\hat{\lambda}^i_{AB} > \lambda^i_{AB}$ and $\hat{f}^i_{AB} > 0$ then $\Delta f^i_{e1} \leq 0$.

c) If $\Delta f_{e1} \geq 0$, $\hat{\lambda}^i_{BA} < \lambda^i_{BA}$ and $f^i_{BA} > 0$ then $\Delta f^i_{e1} \leq 0$.

d) If $\Delta f_{e3} \leq 0$, $\hat{\lambda}^i_{AB} > \lambda^i_{AB}$ and $\hat{f}^i_{AB} > 0$ then $\Delta f^i_{e3} \geq 0$.

e) If $\Delta f_{e3} > 0$, $\Delta f^i_{e3} > 0$ and $f^i_{AB} > 0$ then $\hat{\lambda}^i_{AB} > \lambda^i_{AB}$.

Proof:

a) Assume $\Delta f_{e1} \geq 0$, $\Delta f_{e3} \geq 0$ and $\Delta f^i_{e3} > 0$. By contradiction assume that $\Delta f^i_{e1} > 0$. Then

$$\hat{\lambda}^i_{SD} = K^i_{e1}(\hat{f}^i_{e1}, \hat{f}_{e1}) + K^i_{e3}(\hat{f}^i_{e3}, \hat{f}_{e3}) > K^i_{e1}(f^i_{e1}, f_{e1}) + K^i_{e3}(f^i_{e3}, f_{e3}) \geq \lambda^i_{SD} \tag{31}$$

where the equality sign follows from the observation that $\hat{f}^i_{e1} > f^i_{e1} \geq 0$, $\hat{f}^i_{e3} > f^i_{e3} \geq 0$ together with (6), the first inequality follows from the monotonicity of K, and the second inequality from (6) and (7).
However, using (28) and (29) yields $f^i_{e2} > \hat{f}^i_{e2} \geq 0$, $f^i_{e4} > \hat{f}^i_{e4} \geq 0$ so that

$$\hat{\lambda}^i_{SD} \leq K^i_{e2}(\hat{f}^i_{e2}, \hat{f}_{e2}) + K^i_{e4}(\hat{f}^i_{e4}, \hat{f}_{e4}) < K^i_{e2}(f^i_{e2}, f_{e2}) + K^i_{e4}(f^i_{e4}, f_{e4}) = \lambda^i_{SD} \tag{32}$$

which contradicts (31).

b) Assume $\Delta f_{e1} \geq 0$, $\hat{\lambda}^i_{AB} > \lambda^i_{AB}$ and $\hat{f}^i_{AB} > 0$. Assume by contradiction that $\Delta f^i_{e1} > 0$ (so that $\hat{f}^i_{e1} > 0$). From (28) we conclude that $\Delta f^i_{e2} < 0$ (so that $f^i_{e2} > 0$). Using (8) yields

$$K^i_{e1}(\hat{f}^i_{e1}, \hat{f}_{e1}) + \hat{\lambda}^i_{AB} \leq K^i_{e2}(\hat{f}^i_{e2}, \hat{f}_{e2}) \leq K^i_{e2}(f^i_{e2}, f_{e2}) \leq K^i_{e1}(f^i_{e1}, f_{e1}) + \lambda^i_{AB}. \tag{33}$$

Since $\hat{\lambda}^i_{AB} > \lambda^i_{AB}$ it follows that

$$K^i_{e1}(\hat{f}^i_{e1}, \hat{f}_{e1}) < K^i_{e1}(f^i_{e1}, f_{e1}). \tag{34}$$

On the other hand, since K^i_{e1} is monotone and we assume $\Delta f_{e1} \geq 0$ and $\Delta f^i_{e1} > 0$,

$$K^i_{e1}(\hat{f}^i_{e1}, \hat{f}_{e1}) \geq K^i_{e1}(f^i_{e1}, f_{e1}) \tag{35}$$

hence a contradiction.

c) The proof is very similar to (b): using (28) and repeating the former argument for $e2$ (with A and B interchanged and f and $\hat{f}$ interchanged) will yield $\Delta f^i_{e2} > 0$. Using (28) again yields $\Delta f^i_{e1} < 0$.

d) The claim and its proof are symmetric to (b).

e) Assume $\Delta f_{e3} > 0$, $\Delta f^i_{e3} > 0$ and $f^i_{AB} > 0$. From (29) we conclude that $\Delta f^i_{e4} < 0$ and $\Delta f_{e4} < 0$. So $\hat{f}^i_{e3} > 0$ and $f^i_{e4} > 0$, and using (8) yields

$$\lambda^i_{AB} + K^i_{e4}(f^i_{e4}, f_{e4}) \leq K^i_{e3}(f^i_{e3}, f_{e3}) < K^i_{e3}(\hat{f}^i_{e3}, \hat{f}_{e3}) \leq \hat{\lambda}^i_{AB} + K^i_{e4}(\hat{f}^i_{e4}, \hat{f}_{e4}). \tag{36}$$

But since K is monotone then $K^i_{e4}(\hat{f}^i_{e4}, \hat{f}_{e4}) < K^i_{e1}(f^i_{e4}, f_{e4})$, so that $\hat{\lambda}^i_{AB} > \lambda^i_{AB}$. □

The next lemma establishes the identity of any two NEPs which have equal flows on the parallel link sub-networks (one from A to B and the other from B to A).

Lemma 3 *If f and $\hat{f}$ are two different NEPs, there is at least one user i for which $f^i_{AB} \neq \hat{f}^i_{AB}$ or $f^i_{BA} \neq \hat{f}^i_{BA}$.*

Proof: Assume, to the contrary, that $f^i_{AB} = \hat{f}^i_{AB}$ and $f^i_{BA} = \hat{f}^i_{BA}$ for all i. By (30) it follows that $\Delta f^i_{e1} = \Delta f^i_{e3}$ and $\Delta f_{e1} = \Delta f_{e3}$. Assume further that $\Delta f_{e1} \geq 0$ (the proof is symmetric for $\Delta f_{e1} \leq 0$). Then $\Delta f_{e1} = \Delta f_{e3} \geq 0$. Hence, if f and $\hat{f}$ are different there must be a user i for which $\Delta f^i_{e1} = \Delta f^i_{e3} > 0$. But Lemma 2(a) now implies that $\Delta f^i_{e1} < 0$, a contradiction, so that f and $\hat{f}$ must coincide. □

The following two lemmas exclude the possibility of two distinct NEPs with $f_{e1} \geq \hat{f}_{e1}$.

Lemma 4 *Let f and $\hat{f}$ be two NEPs. If $\Delta f_{e1} \geq 0$ and $\Delta f_{e3} \leq 0$ then $f = \hat{f}$.*

Proof: Assume that f and $\hat{f}$ are distinct. Observing Lemma 3, we may assume that the following properties hold:

$$f^i_{AB} \neq \hat{f}^i_{AB} \text{ for some user } i \tag{37}$$

$$\Delta f_{e1} \geq 0. \tag{38}$$

This entails no loss of generality since A and B can be relabelled if necessary to satisfy (37), and then f and $\hat{f}$ can be interchanged if needed to satisfy (38). Now, there are two possible cases:

a) $\Delta f_{AB} \geq 0$. By (37) it then follows that L^+_{AB} (as defined in (13)) is not empty. Applying Proposition 3 to the parallel links network from A to B, we conclude that there is at least one user i for which $\hat{\lambda}^i_{AB} > \lambda^i_{AB}$ and $\hat{f}^i_{AB} > f^i_{AB} \geq 0$. Furthermore, from Lemma 2(b) $\Delta f^i_{e1} \leq 0$, and from Lemma 2(d) $\Delta f^i_{e3} \geq 0$. Using this and (30) we conclude that $\Delta f^i_{BA} > 0$, so that $\hat{f}^i_{BA} > 0$. But since the NEP cannot contain routing loops (as already explained), that is a contradiction.

b) $\Delta f_{AB} \leq 0$. In this case (30) implies that $\Delta f_{BA} \leq 0$. Furthermore, by (28) and (29) it holds that $\Delta f_{e2} \leq 0$ and $\Delta f_{e4} \geq 0$. This is completely symmetrical to the previous case, and we can repeat the argument above to obtain a contradiction.

It follows that f and $\hat{f}$ cannot be distinct, namely $\hat{f} = f$. □

Lemma 5 *Let f and $\hat{f}$ be two equilibrium points. If $\Delta f_{e1} \geq 0$ then $\Delta f_{e3} \leq 0$.*

Proof: If $f = \hat{f}$ then we are done. Otherwise, it may be assumed as above that (37) holds. Assume further, by contradiction, that $\Delta f_{e1} \geq 0$ and $\Delta f_{e3} > 0$. Consider the following group of users

$$I_1 = \{i \in \mathcal{I} \mid \ \Delta f^i_{e3} > 0\} \tag{39}$$

and the following subsets thereof:

$$\begin{aligned} I_2 &= \{i \in \mathcal{I} \mid \ \Delta f^i_{e3} > 0, f^i_{AB} > 0\} && (40)\\ I'_2 &= \{i \in \mathcal{I} \mid \ \Delta f^i_{e3} > 0, f^i_{AB} = 0\} && (41)\\ I_3 &= \{i \in \mathcal{I} \mid \ \Delta f^i_{e3} > 0, \hat{f}^i_{BA} > 0\} && (42)\\ I'_3 &= \{i \in \mathcal{I} \mid \ \Delta f^i_{e3} > 0, \hat{f}^i_{BA} = 0\}\,. && (43) \end{aligned}$$

Note that $I_1 = I_2 \cup I'_2 = I_3 \cup I'_3$. Recall the definitions of $I^+_{AB} = \{i \in \mathcal{I} \mid \ \hat{\lambda}^i_{AB} > \lambda^i_{AB}\}$ in (13), and similarly define

$$I^-_{BA} = \{i \in \mathcal{I} \mid \ \hat{\lambda}^i_{BA} < \lambda^i_{BA}\} \tag{44}$$

Further define

$$\begin{aligned} I_4 &= \{i \in \mathcal{I} \mid \ \hat{\lambda}^i_{AB} > \lambda^i_{AB}, \hat{f}^i_{AB} > 0\} \subseteq I^+_{AB} && (45)\\ I_5 &= \{i \in \mathcal{I} \mid \ \hat{\lambda}^i_{BA} < \lambda^i_{BA}, f^i_{BA} > 0\} \subseteq I^-_{BA}\,. && (46) \end{aligned}$$

From Lemma 2(a), if $i \in I_1$ then $\Delta f^i_{e1} \leq 0$. According to Lemma 2(b) if $i \in I_4$ then $\Delta f^i_{e1} \leq 0$, and according to Lemma 2(c) if $i \in I_5$ then $\Delta f^i_{e1} \leq 0$. We can now conclude that:

$$\sum_{i \in I_1 \cup I_4 \cup I_5} \Delta f^i_{e1} \leq 0. \tag{47}$$

From Lemma 2(e) if $i \in I_2$ then $i \in I^+_{AB}$, which together with (45) yields

$$I_2 \cup I_4 \subseteq I^+_{AB}. \tag{48}$$

From Proposition 2 it is known that $\sum_{i \in I^+_{AB}} \Delta f^i_{AB} \geq 0$. Moreover by (45), for $i \in I^+_{AB} \setminus (I2 \cup I4)$, it holds that $\hat{f}^i_{AB} = 0$, and therefore $\Delta f^i_{AB} \leq 0$, resulting in

$$\sum_{i \in I_2 \cup I_4} \Delta f^i_{AB} \geq \sum_{i \in I_2 \cup I_4} \Delta f^i_{AB} + \sum_{i \in I^+_{AB} \setminus (I_2 \cup I4)} \Delta f^i_{AB} = \sum_{i \in I^+_{AB}} \Delta f^i_{AB} \geq 0. \tag{49}$$

Notice that from (41) and (46), for $i \in I'_2 \cup I_5$ it holds that $f^i_{AB} = 0$, and therefore $\Delta f^i_{AB} \geq 0$. Combining this with (49) and noting that $I_1 = I_2 \cup I'_2$ gives

$$\sum_{i \in I_1 \cup I_4 \cup I_5} \Delta f^i_{AB} = \sum_{i \in I_2 \cup I'_2 \cup I_4 \cup I_5} \Delta f^i_{AB} \geq 0. \tag{50}$$

Repeating the argument after interchanging A with B and f with $\hat{f}$ yields

$$\sum_{i \in I_1 \cup I_4 \cup I_5} \Delta f^i_{BA} \leq 0. \tag{51}$$

Substituting (50), (51) and (47) in (30) yields

$$\sum_{i \in I_1 \cup I_4 \cup I_5} \Delta f^i_{e3} \leq 0. \tag{52}$$

But by definition of I_1

$$\Delta f_{e3} = \sum_{i \in \mathcal{I}} \Delta f^i_{e3} \leq \sum_{i \in I_1 \cup I_4 \cup I_5} \Delta f^i_{e3} \leq 0 \tag{53}$$

which is a contradiction to the assumption that $\Delta f_{e3} > 0$. □

Lemmas 4 and 5 together prove uniqueness of the equilibrium point in the network in Figure 1(e) when $\Delta f_{e1} \geq 0$. The opposite case of $\Delta f_{e1} \leq 0$ is easily reduced to the previous one by interchanging f and $\hat{f}$. Hence, this completes the proof of Proposition 4.

3.3 Examples for Non-Uniqueness

For general network topologies the NEP need not be unique. In this section we show that a counter-example (namely, a convex network game with multiple Nash equilibria) may be found for any network which is not nearly parallel. Recall from Proposition 1 that in any

network which is not nearly parallel, one of the networks in Figure 2 is embedded in the wide sense. We start with two basic examples:

Example 1: Consider the network in Figure 2(a). Let the cost functions be of the form $J_l^i(f_l^i, f_l) = f_l^i T_l^i(f_l)$, with the latency functions $T_l^i(x)$ and the demand d^i for each user given in the following table

user	d^i	$e1$	$e2$	$e3$	$e4$	$e5$
1	6	"∞"	"∞"	$7x$	x	$f_1(x)$
2	4	"∞"	x	$f_2(x)$	"∞"	$2x$
3	4	$x+21$	"∞"	"∞"	$f_2(x)$	x

where

$$f_1(x) = \begin{cases} x & \text{if } x < 6 \\ \frac{1}{3}\left(e^{3(x-6)} + 17\right) & \text{if } x \geq 6 \end{cases} \tag{54}$$

and

$$f_2(x) = \begin{cases} x & \text{if } x < 4 \\ \frac{1}{3}\left(e^{3(x-4)} + 11\right) & \text{if } x \geq 4 \end{cases} . \tag{55}$$

Note that this functions are continuously differentiable. The infinite values in the table indicate large enough functions, so that the user always prefers not to use this link.

Each user can thus choose to divide its flow between two different routes. User 1 can choose between $e3$ and $e4$–$e5$, user 2 can chose between $e2$–$e5$ and $e3$, and user 3 can choose between $e4$–$e5$ and $e1$. It is easily verified that one Nash equilibrium is formed if each user diverts all of its flow to its first option, and another Nash equilibrium is formed if each user will divert all of its flow to the second option. For example, in the first NEP, (8) may be verified by noting that for user 1,

$$K_{e3}^1(6,6) < K_{e4}^1(0,4) + K_{e5}^1(0,8) \tag{56}$$

and in the second NEP:

$$K_{e4}^1(6,6) + K_{e5}^1(6,6) < K_{e3}^1(0,4). \tag{57}$$

Similar inequalities may be verified for users 2 and 3.

Example 2: A similar example can be constructed for the network in Figure 2(c). The demand and $T_l^i(x)$ functions for each user are:

user	d^i	$e1$	$e2$	$e3$	$e4$	$e5$
1	6	$2x$	"∞"	$2x$	"∞"	$f_1(x)$
2	4	$f_2(x)$	"∞"	"∞"	x	$6x$
3	4	$x+21$	x	$f_2(x)$	"∞"	$6x$

where $f_1(x)$ and $f_2(x)$ are as defined in Example 1. Each user can choose how to divide its flow between link $e5$ and some other route, $e1$–$e3$ for user 1, $e1$–$e4$ for user 2 and $e2$–$e5$ for user 3. It may be verified as above that one Nash equilibrium is formed when user 1 transfers all its flow through $e5$, and users 2 and 3 choose not to use $e5$ at all. Another Nash equilibrium is formed when users 2 and 3 ship all their demand on $e5$, and user 1 does not use $e5$ at all.

These two examples show that multiple equilibria exist in the networks of Figure 2(a) and 2(c). We need now to extend the examples to the other networks in Figure 2, and then to any network in which these basic networks are embedded. To this end, we will require the considered equilibrium point to be stable with respect to small perturbations, so that the series addition of links with small enough cost does not alter the equilibrium. We use the following definition.

Definition 6 *A Nash equilibrium of the network game is called* strong *if for any path p from O to D that is used by user i, namely $f_l^i > 0$ for every $l \in p$, it holds that*

$$\sum_{l\in p} K_l^i(f_l^i, f_l) < \sum_{l\in p'} K_l^i(f_l^i, f_l) \tag{58}$$

for any other path p' that connects O and D.

Note that in a strong NEP each user employs a unique path from origin to destination. It may be verified that the NEPs in examples 1 and 2 are strong.

Lemma 6 *Let $\mathcal{G}$ be a network over which there exists a convex network game with two distinct strong NEPs. Then for any network $\mathcal{G}'$ in which $\mathcal{G}$ is embedded in the wide sense, there exists a convex network game with two different strong NEPs.*

Proof: The proof is similar to that of a corresponding claim in [20]. Let f and $\hat{f}$ be the two strong NEPs in $\mathcal{G}$ and denote by p^i and $\hat{p}^i$ the unique paths of user i in f and $\hat{f}$ respectively. Since the definition of embedding in the wide sense is recursive we need only consider the case

where $\mathcal{G}'$ was obtained from $\mathcal{G}$ by one of the following operations: (1) the subdivision of an edge, (2) the addition of an edge, or (3) the subdivision of a terminal vertex. In case (1) the cost function of each direction of the edge that was subdivided is equally split between its two parts. It is trivially seen that the new game over $\mathcal{G}'$ remains a convex network game, which supports the two distinct equilibria of $\mathcal{G}$. In case (2) we may set the cost functions of each user on the added edge so that $K_l^i(0,0)$ is higher then $\max(\sum_{l\in p^i} K_l^i(f_l^i, f_l), \sum_{l\in \hat{p}^i} K_l^i(\hat{f}_l^i, \hat{f}_l))$. In that case no user has an incentive to use the new edge and the equilibrium points do not change. In case (3) a new node is added to $\mathcal{G}$, with an edge that connects it either to D or to O. If we would set the cost on that link to zero the NEPs would obviously not be affected. However, since a null cost violates Assumption A3, we choose a small non-zero cost function for that link. Since the two equilibria in $\mathcal{G}$ are strong we may set the marginal costs $K_l^i(f_l^i, f_l)$ on that link sufficiently small f and $\hat{f}$ are still NEPs. Specifically we select J_l^i so that $K_l^i(f_l^i, f_l)$ is less then the difference between used routes and unused routes for $f_l^i \leq d^i$. □

Proposition 5 *For every network $\mathcal{G}$ in which one of the networks in Figure 2 is embedded in the wide sense, one can find a convex network game for which the equilibrium is not unique.*

Proof: Example 1 demonstrates the claim for the network in Figure 2(a). A symmetric example can be used for the network in Figure 2(b). Example 2 shows the same for the network in Figure 2(c). We can apply Example 2 to the network in Figure 2(d) by imposing small enough costs on the additional link $e6$ without affecting the two strong equilibria (as argued in Lemma 6). Lemma 6 now proves the proposition. □

Propositions 4 and 5 together with Proposition 1 provide the proof of Theorem 1.

Remark: Consider a directional network, where the directional links do not necessarily come in pairs as we have assumed so far. Obviously, if the network topology corresponds to a nearly parallel network, then the sufficiency part of Theorem 1 (uniqueness of the Nash equilibrium) still holds. Indeed, any "missing" link which is required to complete a pair may be added with large enough costs. As for the necessity part of Theorem 1, we need to be more careful here in the definition of networks that are not nearly-parallel. For example, in network (a) of Figure 2, suppose that link $e2$ has opposite directionality to that of $e5$. Then $e2$ can be effectively eliminated, and the network reduces to a parallel link network. A similar reduction occurs in network (c) if $e1$ and $e3$ share the same directionality while $e2$ and $e4$ have the opposite one (see [20]). Without formalizing this idea, we conclude that ineffective paths should be purged before the network can be properly classified.

4 Weakly Convex Network Games

In this section we shall consider the uniqueness of the Nash equilibrium under slightly weaker conditions on the cost functions. Under these weaker conditions, it will be possible to consider the *Wardrop* equilibrium (for a finite number of user classes) as a special case of our model. We start by delineating the relation between the Wardrop and Nash equilibria for the network model.

4.1 Wardrop and Nash Equilibria

Consider the network model as defined above, where now $i \in I$ refers to the *user class*. The latency of link l for users of class i is given by the function $T_l^i(f_l)$, which we assume to be strictly increasing. A flow pattern f is a Wardrop equilibrium if

$$\sum_{l \in p^i} T_l^i(f_l) = \min_p \sum_{l \in p} T_l^i(f_l) \quad \text{for every } i \in I \tag{59}$$

where p^i is any route employed by user i, and p' is any other feasible route for user i.

It has been established in [20] that the Wardrop equilibrium is unique for any choice of (monotone) latency functions $T_l^i(f_l)$ if and and only if the network is nearly parallel. [1]

The Wardrop equilibrium and the Nash equilibrium finitely-many users may be related from two different viewpoints. First, the Wardrop equilibrium may be obtained as the limit of the Nash equilibrium point when the number of users is increased to infinity while their flow demand is reduced accordingly [13]. More relevant here, the Wardrop equilibrium is mathematically equivalent to a Nash equilibrium with properly defined costs, as we indicate below. This relation was already observed in [5] for the Wardrop equilibrium with a single user class, which is equivalent to a (single-user) optimization problem; see also [9]. Later, [10] indicated the equivalence of a (still single-class) Wardrop equilibrium to a Nash equilibrium problem with multiple players, where each player stands for a different origin-destination pair.

Starting with the model for multi-class Wardrop model with a finite number of user classes (indexed by i) and link costs $T_l^i(f_l)$, consider the game model where player i corresponds to

[1] Milchtaich's model actually allows for an infinite number, or a continuum, of user classes. We do not consider this possibility here.

user class i, and the link costs are defined by

$$J_l^i(f_l^i, f_l) \stackrel{\triangle}{=} \int_0^{f_l^i} T_l^i(f_l - f_l^i + x)dx\,. \tag{60}$$

It is easily seen that T_l^i is the marginal cost for this cost function, namely $K_l^i(f_l^i, f_l) = T_l^i(f_l)$. As T_l^i is assumed to be a strictly increasing function, it follows that the cost of each player is strictly convex in its own decision variables. The equivalence between the Nash equilibrium for this game and the Wardrop equilibrium that we started with follows immediately by comparing the optimality conditions for the Nash equilibrium in (8) with the definition of the Wardrop equilibrium above.

It is readily seen that the cost functions in 60 satisfy our basic Assumptions A1-A3, *except* for the fact that K_l^i is not strictly increasing in f_l^i (in fact, it only depends on f_l), which violates Assumption A3. This motivates us to study the uniqueness of the Nash equilibrium under somewhat weaker assumptions.

4.2 Uniqueness of the NEP for Weakly Convex Games

Consider the following weaker version of Assumption A3.

Assumption A3': Same as Assumption A3, except that the cost functions $K_l^i(f_l^i, f_l)$ are only required to be *non-decreasing* in f_l^i.

We define a *weakly convex network game* as a convex network game with Assumption A3 replaced by Assumption A3'. The following uniqueness property may be established for this model.

Theorem 2 *If $\mathcal{G}$ is nearly parallel, then the link flows are the same in all Nash equilibrium points for every weakly convex network game $(I, d, \mathcal{J})$ over $\mathcal{G}$.*

Remarks:

1. While the equilibrium link flows are unique, uniqueness need not hold for the per-user flows in weakly convex games. This may be easily realized for the corresponding Wardrop equilibrium, where users with similar latency functions can simply switch links. Uniqueness of the Wardrop equilibrium is however a generic property in the space of latency

functions [20]. A similar property is likely to hold for the present model, but we shall not pursue this point here.

2. It follows from the equivalence outlined above that this result implies uniqueness of the link flows for the Wardrop equilibrium, as defined in (59). In fact, it generalizes the latter by allowing link latencies $T_l^i(f_l^i, f_l)$ to depend on the user flows f_l^i, provided that they satisfy the monotonicity requirements in Assumption A3'.

We may now proceed with the proof of Theorem 2.

Lemma 7 *Lammas 1, 2 and 5, continue to hold for weakly convex games, with the following modifications:*

(i) In Lemma 1, we require that one of the inequalities $\hat{f}_l \geq f_l$ or $\hat{\lambda}^i \leq \lambda^i$ would be strict.

(ii) In Lemma 2(a), we require $\Delta f_{e3} > 0$ instead of $\Delta f_{e3} \geq 0$.

Proof: The same as the original proof of the lemmas with the strict inequalities removing the need for strict monotonicity. □

In light of these modifications, Proposition 3 needs to be refined.

Lemma 8 *Consider a weak convex game $(I, d, \mathcal{J})$ over a parallel link network from A to B. If $\sum_{i\in\mathcal{I}} \hat{f}_{AB}^i \geq \sum_{i\in\mathcal{I}} f_{AB}^i$ and L_{AB}^+ is not empty, then there is at least one user i for which $\hat{f}_{AB}^i > f_{AB}^i$ and $\hat{\lambda}^i > \lambda^i$.*

Proof: Proposition 2 still holds. Therefore, if L_{AB}^+ is not empty then (22) holds with a strict inequality, so that there exist at least one user i for which $\hat{f}_{AB}^i > f_{AB}^i$ and $\hat{\lambda}^i > \lambda^i$. □

Lemma 3 also needs to be sharpened.

Lemma 9 *If f and $\hat{f}$ are two NEPs such that there is a link $l \in \mathcal{L}$ for which $\hat{f}_l \neq f_l$, then there is at least one link $l \in L_{AB} \cup L_{BA}$ for which $\hat{f}_l \neq f_l$.*

Proof: Assume by contradiction that $\hat{f}_l = f_l$ for every $l \in L_{AB} \cup L_{BA}$. Since there is an $l \in \mathcal{L}$ for which $\hat{f}_l \neq f_l$, we can assume without loss of generality that $\Delta f_{e1} > 0$ (relabelling

O,D and A,B if necessary). From (30), $\Delta f_{e3} > 0$. Lemma 5 now leads to the desired contradiction. □

In light of Lemma 9 and Lemma 5 it is sufficient to show that the existence of two equilibria that satisfy the following conditions leads to a contradiction:

$$f_l \neq \hat{f}_l \text{ for some link } l \in L_{AB} \tag{61}$$

$$\Delta f_{e1} \geq 0 \tag{62}$$

$$\Delta f_{e3} \leq 0\,. \tag{63}$$

This is valid since A and B can be relabelled if necessary and then f and $\hat{f}$ can be interchanged if needed.

Lemma 10 *Let f and $\hat{f}$ be two equilibrium points. If $\Delta f_{e3} \leq 0$ and $\Delta f_{e1} \geq 0$, then $f_l = \hat{f}_l$ for every $l \in \mathcal{L}$.*

Proof: The proof is very similar to the proof of Lemma 4, using Lemma 8 instead of Proposition 3. □

Lemma 10 together with Lemma 5 establish the claim of Theorem 2.

5 Conclusion

As networks become larger and less centralized, it is usually hard to give theoretical predictions regarding the precise operating conditions of the network. Equilibrium analysis provides a useful proxy for this purpose, which has been used both for the qualitative understanding of basic phenomena, as well as for setting up the quantitative models that are essential for network management. Uniqueness of the equilibrium is important both for analysis and management. When the equilibrium is not unique, the network behavior becomes less predictable. Simulation results, for example, cannot be relied on to give a complete picture of the network operation. From the management point of view, several schemes have propose the use of pricing and related management tools to enforce the efficiency of the equilibrium. It is usually easy to maintain the efficiency of some equilibrium point, but when the equilibrium is not unique it is not clear that the intended equilibrium point will indeed be the one to take effect, and the management task tends to become much harder.

This paper provides a complete characterization of two-terminal network topologies for which the Nash equilibrium is unique, under broad conditions on the cost functions, and for any number and size of network users. Unfortunately, the class of networks for which this broad sense of uniqueness holds is somewhat restricted. Thus, alongside the verification of uniqueness for nearly parallel networks, the result also points out those network configuration that might bring about multiple equilibria.

We have not dealt in this paper with multi-terminal networks, in the sense that different flows (of different users, or even of the same user) may correspond to different source and destination pairs. While either necessary or sufficient conditions may be extracted from our results, it remains open whether a complete characterization of topological uniqueness may be given for this case.

When uniqueness cannot be inferred from the network topology alone, further properties of the cost functions and user characteristics clearly come into play. The well known diagonal strict convexity conditions [13, 22] on the cost functions are often restrictive, and one may hope to find a middle ground that combines cost function properties with other network and user characteristics. This remains an interesting direction for further research.

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